Practical 3

Ex.1 Water Samples with a Pollutant (Binomial)

Items	Value
Number of samples (n)	18
Probability a sample has the pollutant (p)	0.1
q = 1 − p	0.90

*Find the probability	P()	value	Equation	Answer
(a) Exactly 2 samples contain the pollutant	P(X = x)	2	P(X=x) = C(n,x) · p^x · q^(n−x)	**0.2835**
(b) At least 4 samples contain the pollutant	P(X ≥ x)	4	P(X≥x) = 1 − P(X≤x−1)	**0.0982**

Ex.2 Robot Reporter Stories (Binomial)

Items	Value
Number of stories (n)	15
Proportion written by the algorithm (p)	0.65
q = 1 − p	0.35

*Find the probability	P()	value	Equation	Answer
(a) Exactly 11 stories written by the algorithm	P(X = x)	11	P(X=x) = C(n,x) · p^x · q^(n−x)	**0.1792**
(b) At least 10 stories written by the algorithm	P(X ≥ x)	10	P(X≥x) = 1 − P(X≤x−1)	**0.5643**
(c) Between 8 and 11 inclusive	x_1 (lower)	8	$P(x_1 \le X \le x_2) = P(X \le x_2) - P(X \le x_1 - 1)$	**0.7141**
	x_2 (upper)	11		

Ex.3 Flaws in Thin Copper Wire (Poisson)

Items	Value
Mean number of flaws per millimetre (rate)	2.3
(a) Length of wire, L (mm)	5
(a) Mean flaws in this length, λ = rate × L	11.50
(b) Length of wire, L (mm)	2
(b) Mean flaws in this length, λ = rate × L	4.60

*Find the probability	P()	value	Equation	Answer
(a) Exactly 10 flaws in 5 mm of wire	P(X = x)	10	P(X=x) = e^(−λ) · λ^x / x!	**0.1129**
(b) At least one flaw in 2 mm of wire	P(X ≥ x)	1	P(X≥x) = 1 − P(X≤x−1)	**0.9899**

Ex.4 Alpha Particles from Radon Gas (Poisson)

Items	Value
Mean number of particles next week (λ)	1.3

*Find the probability	P()	value	Equation	Answer
(a) Exactly one particle next week	P(X = x)	1	P(X=x) = e^(−λ) · λ^x / x!	**0.3543**
(b) One or more particles next week	P(X ≥ x)	1	P(X≥x) = 1 − P(X≤x−1)	**0.7275**
(c) At least two but no more than four	x_1 (lower)	2	$P(x_1 \le X \le x_2) = P(X \le x_2) - P(X \le x_1 - 1)$	**0.3625**
	x_2 (upper)	4		

Ex.5 Defective Book Bindings (Binomial vs Poisson Approximation)

Items	Value
Number of books bound (n)	100
Probability a binding is defective (p)	0.05
λ = n·p (for the Poisson approximation)	5.00

*Find the probability	P()	value	Equation	Answer
(a) Exactly 2 defective, binomial formula	P(X = x)	2	P(X=x) = C(n,x) · p^x · q^(n−x)	**0.0812**
(b) Exactly 2 defective, Poisson approximation	P(X = x)	2	P(X=x) ≈ e^(−np) · (np)^x / x!	**0.0842**